

CORUGEA, Romania

WMO: 154080

Lat: 44.7344N

Lon: 28.3419E

Elev: 220

StdP: 98.71

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.7	-14.9	1.1	-8.8	-12.5	1.3	-8.4	16.2	-5.6	15.4	-4.6	4.1	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	31.7	20.4	30.4	20.0	29.1	19.6	23.5	27.7	22.3	26.7	21.3	25.8	4.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.7	17.9	23.9	21.1	16.2	23.4	20.0	15.1	22.8	71.3	27.9	66.6	26.6	62.9	25.6	32.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.2	9.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-2.0	1.6	5.0	9.8	16.2	21.0	22.4	22.4	17.5	11.5	6.6	1.2
	DBStd	9.27	4.47	5.00	4.62	4.04	3.35	2.96	2.25	3.32	4.16	3.93	4.62	4.32
	HDD10.0	1251	372	238	163	52	2	0	0	0	31	119	274	
	HDD18.3	3027	630	469	412	257	82	9	1	5	66	213	351	532
	CDD10.0	1669	0	2	9	46	194	329	383	384	226	76	18	1
	CDD18.3	405	0	0	0	1	16	88	126	131	42	1	0	0
	CDH23.3	3368	0	0	1	18	127	694	1043	1163	317	4	0	0
	CDH26.7	1052	0	0	0	3	21	191	326	431	82	0	0	0
(14) Wind	WSAvg	3.8	4.0	3.9	4.4	4.7	4.0	3.5	3.1	3.2	3.5	3.7	3.8	3.9
(15) Precipitation	PrecAvg	443	28	17	28	32	49	54	50	40	46	34	31	33
	PrecMax	647	81	48	72	79	121	108	128	115	105	68	73	89
	PrecMin	259	3	2	0	2	9	13	9	5	2	2	0	3
	PrecStd	92	20	12	17	17	28	24	30	30	33	17	19	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	16.9	21.1	25.7	28.8	31.9	33.2	33.2	30.9	23.8	20.1	12.4
		MCWB	8.2	12.7	12.6	16.6	19.1	21.9	21.5	20.8	20.6	16.7	15.0	10.6
	2%	DB	7.3	13.0	17.3	22.0	26.4	30.4	31.4	32.0	28.9	21.8	16.8	10.4
		MCWB	6.0	9.6	10.6	14.3	18.4	20.8	19.9	20.4	17.9	15.2	14.1	9.5
	5%	DB	5.4	10.2	14.5	19.3	24.4	28.8	29.7	30.7	26.7	19.6	14.5	8.4
		MCWB	4.8	7.9	9.4	13.0	16.8	20.5	19.6	20.0	17.4	14.6	12.3	7.3
	10%	DB	4.1	7.8	11.9	17.1	22.8	27.2	28.4	29.1	24.6	17.3	12.7	6.6
		MCWB	3.6	6.3	7.7	11.8	16.1	19.4	19.2	19.7	16.9	13.1	11.2	5.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	13.0	13.4	17.3	21.8	24.1	24.4	29.9	26.1	18.9	17.1	11.7
		MCDB	9.3	14.8	19.5	24.4	26.0	29.0	27.4	29.9	26.1	20.3	18.0	12.3
	2%	WB	6.4	10.1	11.5	15.5	19.8	22.8	22.8	23.7	20.6	16.7	14.6	9.8
		MCDB	7.2	12.7	15.9	20.8	24.1	27.5	26.9	28.0	23.4	19.9	15.9	10.2
	5%	WB	4.9	7.9	10.0	13.9	18.4	21.7	21.8	22.0	19.2	15.4	12.9	7.7
		MCDB	5.6	9.9	13.8	18.3	22.7	26.3	26.0	26.9	22.7	18.0	14.5	8.6
	10%	WB	3.7	6.3	8.3	12.3	17.3	20.8	20.8	20.8	18.1	14.2	11.4	6.0
		MCDB	4.2	7.6	11.2	16.4	21.4	25.5	25.1	26.0	23.2	16.5	12.9	6.6
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.7	8.7	10.0	10.0	10.5	10.7	10.4	10.1	8.5	6.9	5.5
		MCDBR	7.1	10.7	13.7	13.9	12.3	12.3	12.9	11.9	12.7	11.5	9.6	7.7
	5% WB	MCWBR	6.0	8.0	8.0	7.6	6.1	5.6	5.4	5.8	5.7	6.0	6.7	6.5
		MCWBR	6.7	9.8	11.8	12.5	11.2	11.4	11.2	10.9	10.6	10.5	9.0	7.9
(40) Clear-Sky Solar Irradiance	taub		0.335	0.362	0.394	0.428	0.407	0.408	0.423	0.432	0.404	0.377	0.355	0.335
		taud	2.322	2.260	2.218	2.143	2.283	2.305	2.274	2.239	2.290	2.343	2.352	2.347
	Ebn at Noon		763	807	831	832	864	862	844	820	812	782	733	721
		Edh at Noon	81	103	123	144	129	127	130	130	114	94	79	73
(44) All-Sky Solar Radiation	RadAvg		1.28	2.09	3.31	4.75	6.00	6.48	6.49	5.81	4.22	2.66	1.47	1.11
	RadStd		0.19	0.29	0.33	0.47	0.38	0.38	0.35	0.32	0.32	0.26	0.19	0.16

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-122	+89	+57

Nomenclature: See separate page